

# Vishnu Sundhan

Software Engineer | GenAI & Agentic Systems | Applied AI  
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Applied AI Engineer who starts from an ambiguous problem and learns the domain on the job, with over a year across 4 internships in RAG, agent workflows, evaluation, and Python backends, and current NYU research on agent and LLM observability. 7 competition wins for rapidly prototyping microservices and multi-agent systems, including the MIT Hackathon, Kaggle ML Olympiad, Montclair Hack, 3 Percent Hacks, and Stevens Hackathon.

## EDUCATION

### New York University

M.S. in Computer Science | GPA: 3.78

New York, USA  
Aug 2024 – May 2026

### Rajalakshmi Engineering College, Anna University

B.E. in Computer Science and Engineering | GPA: 8.94/10

Chennai, India  
Nov 2020 – May 2024

## EXPERIENCE

### Runova Inc

Software Engineer Intern

New York, USA  
Jan 2026 – May 2026

- Developed a fully offline Android skin-scan app in Kotlin by moving a Flask REST API onto the phone. Trained an EfficientNet on about 6,000 public images, balanced across dry, normal, and oily skin and augmented, then converted it to float16 TFLite at about 87% accuracy on held-out images.
- CameraX and coroutines run BlazeFace and that model in parallel on half-resolution scans, with crop-offset overlays, cutting on-device analysis from about 15 seconds to under 3 seconds. Then built Prelude, a relationship assistant, as a Netlify web app and Chrome extension with Supabase memory.
- Users paste a chat, set the relationship stage and tone, and get a boundary-checked reply. Compared replies with GPT-4, Claude, and Gemini on 500 scenarios and seven criteria, logging token cost and latency per call, judged by Gemini 2.5 Flash Lite with Evidently checks, before end-user release.

### Tata Technologies

Software Engineer Intern (Gen AI)

Pune, India  
Jun 2025 – Jul 2025

- Built a multi-agent service with LangGraph and CrewAI on FastAPI and a React/TypeScript UI for project timeline coordination and a meeting assistant that called tools in a ReAct loop. Microsoft Presidio scrubbed personal details before text was stored, cutting one planning cycle by 2 weeks.
- On a tube mill, laser-triangulation sensors traced roundness as flat strip rolled into pipe. Compared isolation forest, one-class SVM, ARIMA, etc. on that time series, then trained a PyTorch net from scratch. A dashboard cached repeat diagnostic queries, and the chosen models reached 96% precision.

### Populus Empowerment Network

Python Data Science Intern

Chennai, India  
Jan 2024 – Jun 2024

- Worked on fault-tolerant ETL pipelines from ingesting various web sources and cleaning data to dashboard visualizations for processing 62M+ bilingual electoral records using Python, MySQL, and Selenium. Maintained 90%+ data consistency and reliable downstream analytics workflows for stakeholders.
- Automated data extraction from ambiguous scanned documents using Python, a CNN, coordinate correction, Tesseract OCR, and Amazon Textract, which cut manual retyping and extraction errors across 10,000+ pages per Assembly Constituency, plus bilingual sentiment on YouTube transcripts.

### Rajalakshmi Engineering College

AI Research Assistant (6 published papers: 5 IEEE, 1 SSRN)

Chennai, India  
Oct 2023 – May 2024

- Co-authored an IEEE paper and a Flask web app that classifies brain MRI into glioma, meningioma, pituitary, or no tumor with a CNN and a PCA plus RBF-SVM, showing thresholding and contours. On 7,023 images the models reached 97% on pituitary and no-tumor and 91% on meningioma.
- Co-authored an IEEE paper comparing EfficientNet B0, B1, and B3 using transfer learning on 6,400 2D ADNI MRI slices across four dementia classes, with Grad-CAM maps of the region behind each prediction. EfficientNet B3 reached 99.06% accuracy, with precision 0.992 and recall 0.989.

### LogicField Technologies Pvt Ltd

Machine Learning Intern

Chennai, India  
Jun 2023 – Aug 2023

- Developed a geospatial risk assessment ensemble model (XGBoost, LightGBM) in Python utilizing GeoPandas and MLflow for optimizing insurance taxes, processing over 5000 historical records to accurately forecast location-specific disaster probabilities (earthquakes, tsunamis, etc) with 89% accuracy.
- Conducted comprehensive data analysis to provide actionable recommendations that reduced potential financial exposure for enterprise clients. Designed and deployed an interactive Node.js and Express dashboard to serve real-time risk metrics, replacing manual report assembly for analysts.

## PROJECTS

### The Bitcoin Quorum - Autonomous Multisig AI Treasury

Apr 2026

- Developed multi-agent autonomous Bitcoin treasury using LangGraph with 3-of-4 Miniscript multisig enforcement; agents independently think, vote, and cryptographically co-sign, making unilateral fund movement mathematically impossible – MIT Bitcoin Hackathon Community Pool Winner.
- Orchestrated five parallel fraud checks: Tavily vendor search, a 30-day duplicate scan, Bitcoin Abuse blacklists, WHOIS for domains under six months, and Blockstream address history. ChromaDB RAG supplied live policy, and Vapi took real-time voice treasury commands on a Next.js dashboard.

### Wound Route - Medicare wound claim router

Jun 2026

- Ingested 300 synthetic skilled-nursing patients in Python from a mock clinical API into SQLite, then routed each wound claim through Medicare Part B rules. Async fetching with aiohttp finished in about 47 seconds versus about 8 minutes on threads, and honored each Retry-After on 429 responses.
- Extracted wound measurements with regex on assessments first and notes second, then applied a 3-step gate: active coverage, complete documentation, and cross-source agreement. Gemini filled about 14 gaps, kept every regex hit, and a traffic light React dashboard listed each patient's route.

### Clarify - Spatial AR product support

Apr 2026

- Built a Next.js support platform where a manufacturer labels hotspots on a 3D product, publishes a phone WebXR view by QR, and grounds a voice and text assistant in that product's PDF manual and custom Q&A. Analytics turns repeated confusion into a spatial heatmap for support teams.

### VisaDraft Studio - Offline USCIS form drafts

Dec 2025

- Built an offline Python desktop app with MRZ and pdfplumber that extracts a passport, I-94, and I-20 with Tesseract, maps the fields onto fillable USCIS forms I-765 and I-539, and runs a pre-flight checklist before exporting the draft. Gemini plus Chroma answers only from official instruction PDFs.

## SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, R, Kotlin, HTML/CSS

**Backend & APIs:** DSPy, JEV, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, Node.js, React.js, LangGraph, CrewAI, LangChain, Google ADK, AutoGen, PydanticAI, Docker, Git, Linux, CI/CD, Caching, System Design, Redis, Ollama, Selenium, Presidio

**Frontend & Big Data:** Next.js, PowerBI, Pandas, GeoPandas, NumPy, Matplotlib, Plotly, Dash, Jupyter, D3.js, GraphRAG, Hadoop, Apache Spark

**ML & MLOps:** PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, Hugging Face, Transformers, MLflow, Prompt Engineering, Fine-Tuning, NLP

**Cloud & Databases:** GCP, BigQuery, Vertex AI, AWS SageMaker, AWS S3, DynamoDB, ChromaDB, Gemini API, Amazon Textract, Tesseract